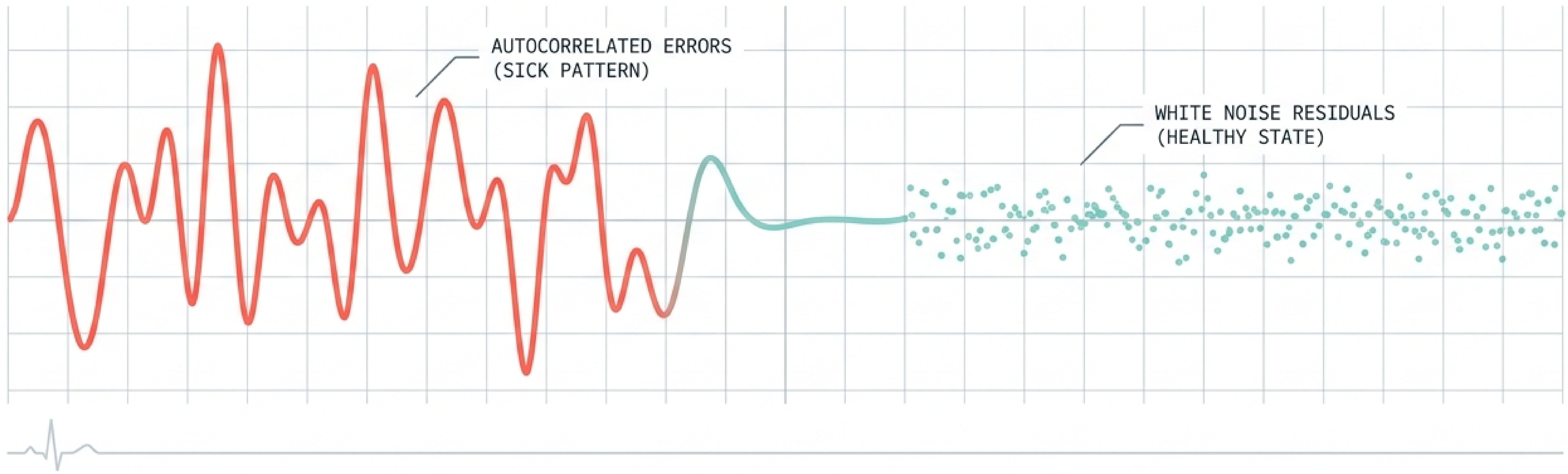
[STATUS: SYSTEM DIAGNOSTIC READY]

[TARGET AUDIENCE: BACHELOR OF ECONOMETRICS]

The Autocorrelation Anomaly

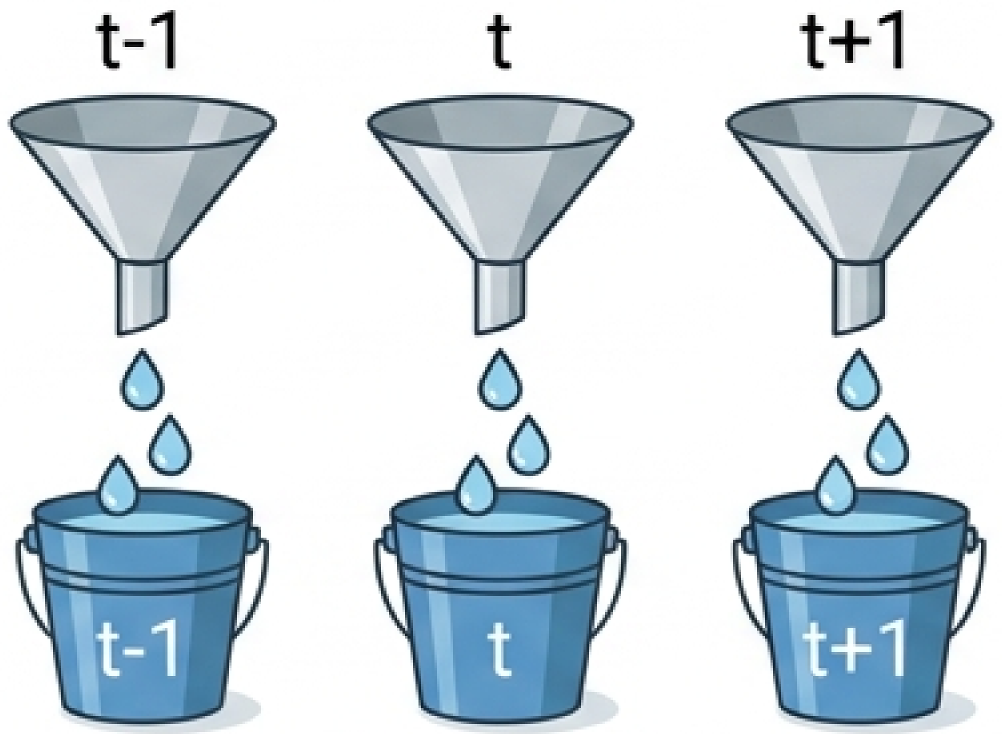
Diagnosing and Curing Correlated Errors in Time Series Data.



NotebookLM

What is Autocorrelation?

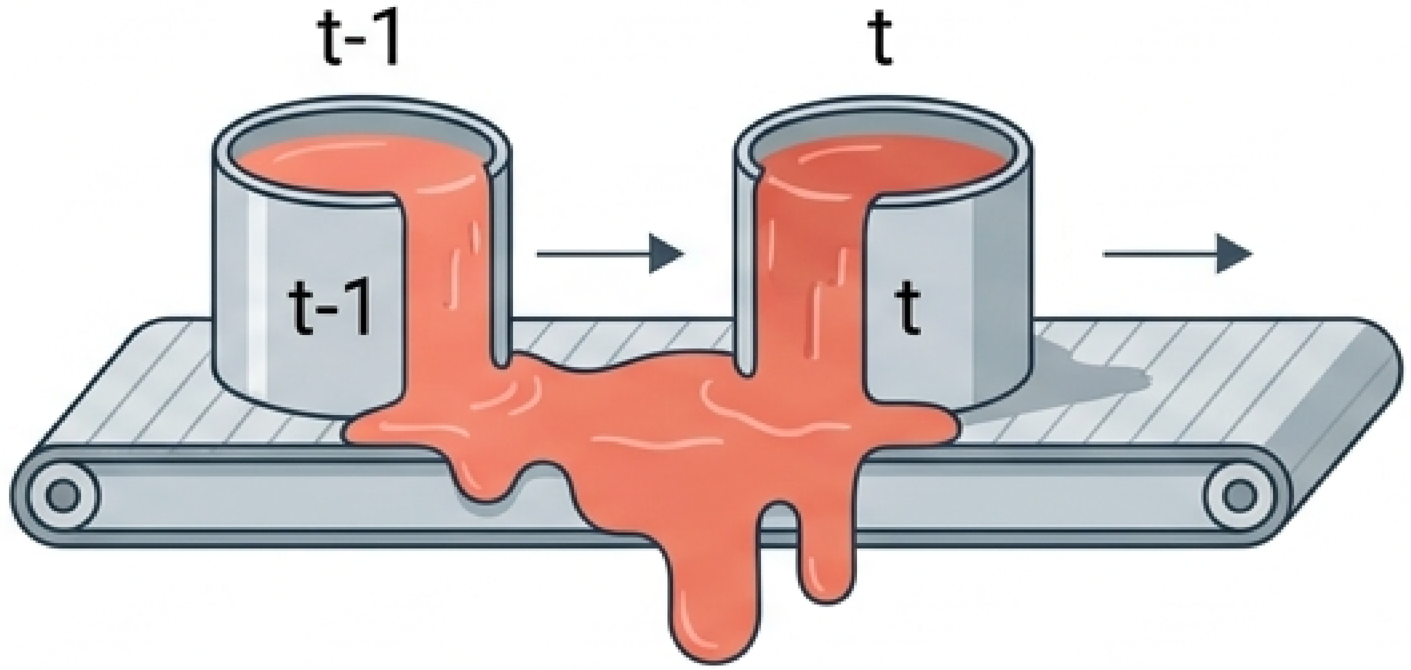
The Healthy Model: CLRM Assumption



$$\text{cov}(u_i, u_j) = E(u_i u_j) = 0 \quad (i \neq j)$$

What happens yesterday stays in yesterday.

The Sick Model: AR(1) Time-Travel Effect



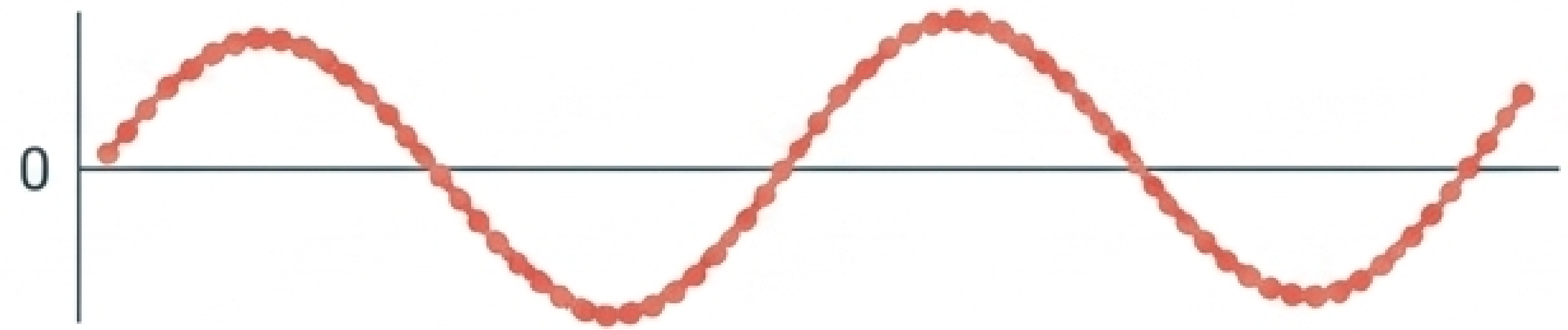
$$u_t = \rho\, u_{t-1} + \varepsilon_t$$

ρ = coefficient of autocovariance. ε_t = pure white noise.

Yesterday's disruption bleeds into today.

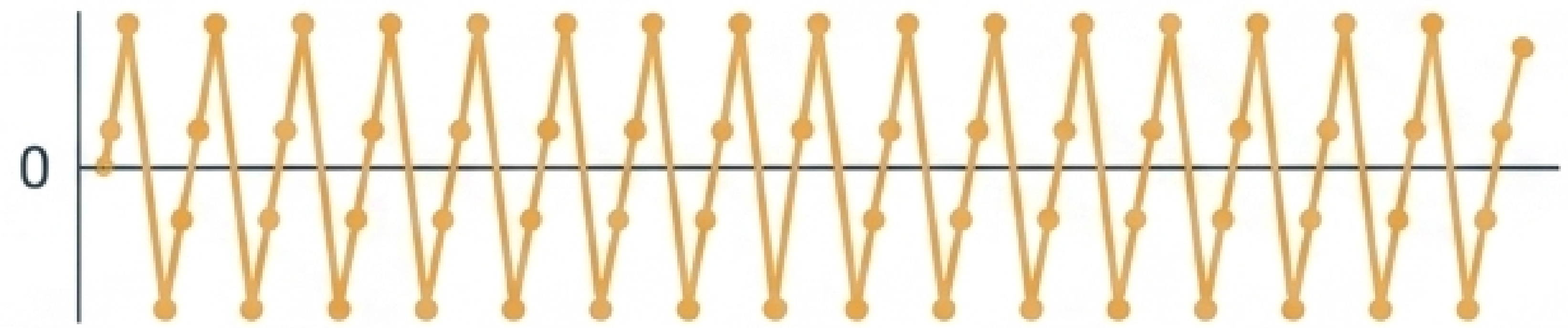
Visualizing the Symptoms

Positive Autocorrelation



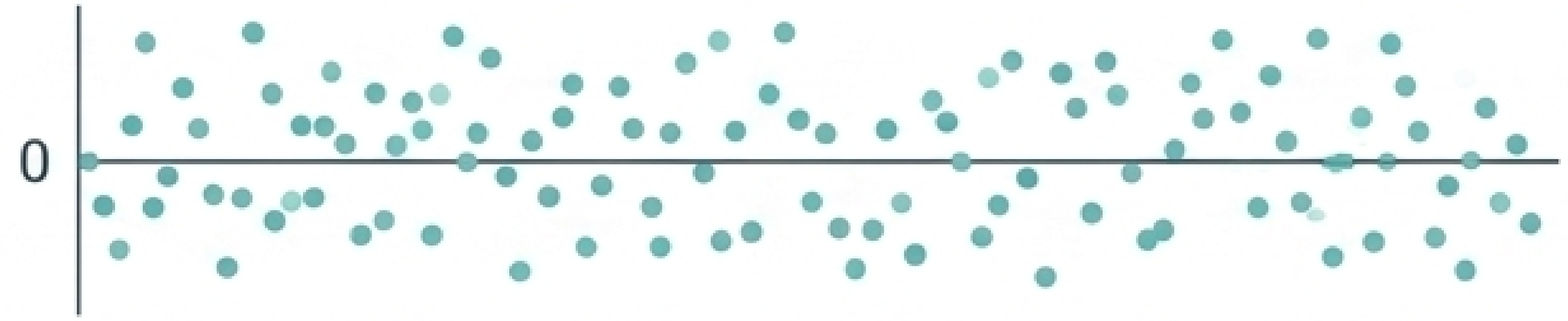
Most common in economics. Inertia keeps values moving together. (e.g., GDP momentum)

Negative Autocorrelation



A continuous over-correction. Less common, often induced by data manipulation.

Nonautocorrelation (Healthy)

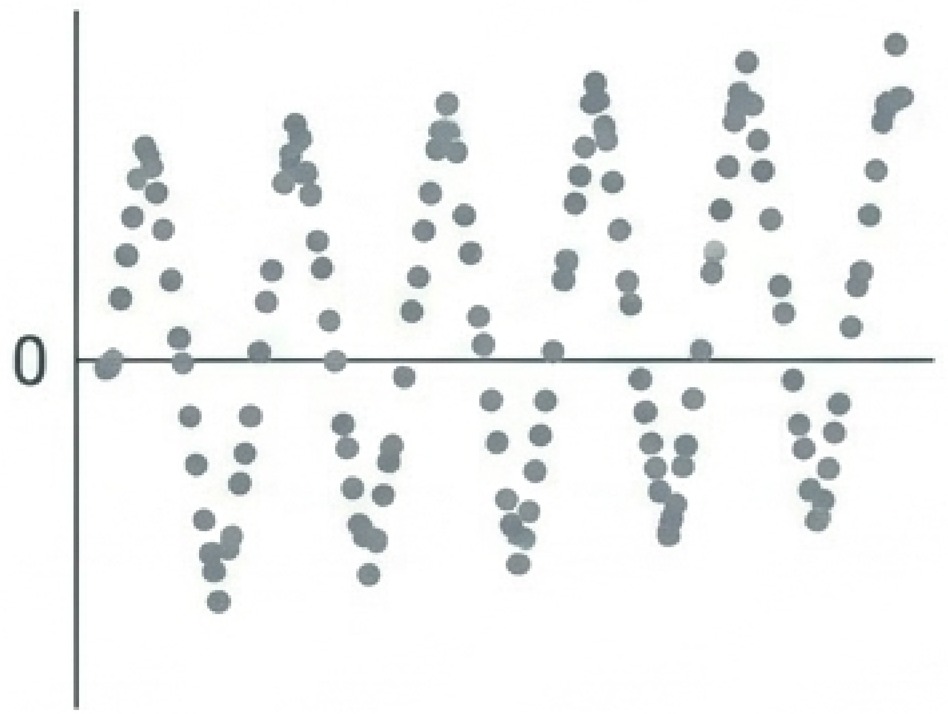


The Classical Linear Regression Model ideal. Pure white noise.

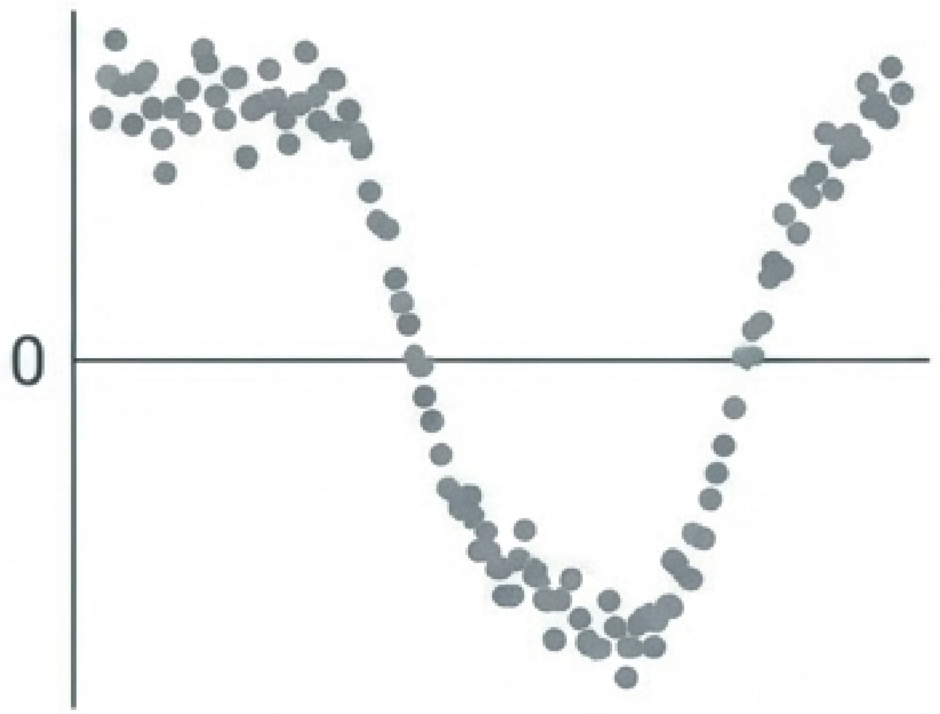
Clinical Rotation: Spot the Sickness

Class Work: Match the patient to the diagnosis. Which plot shows Positive Autocorrelation, Negative Autocorrelation, or Healthy White Noise?

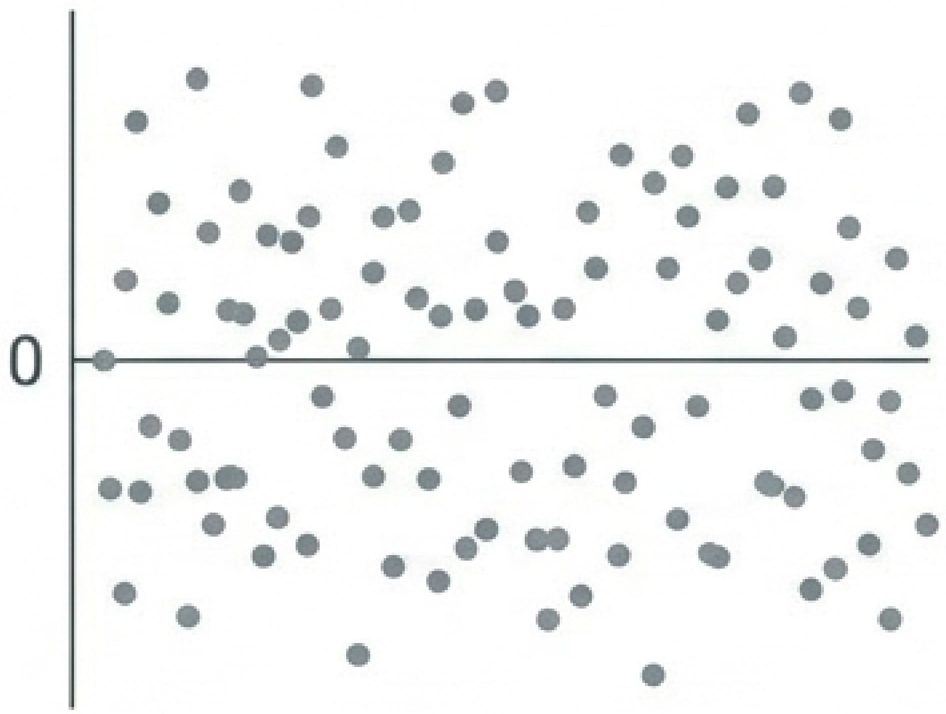
Patient A



Patient B



Patient C



[Answers - A: Negative, B: Positive, C: Healthy]

Root Causes: Why Models Get Sick

Root Causes of Autocorrelation

The Nature of the World

- **Inertia**: Macroeconomic sluggishness (business cycles, GDP momentum).
- **Cobweb Phenomenon**: Agricultural supply reacting to previous year's prices with a lag.

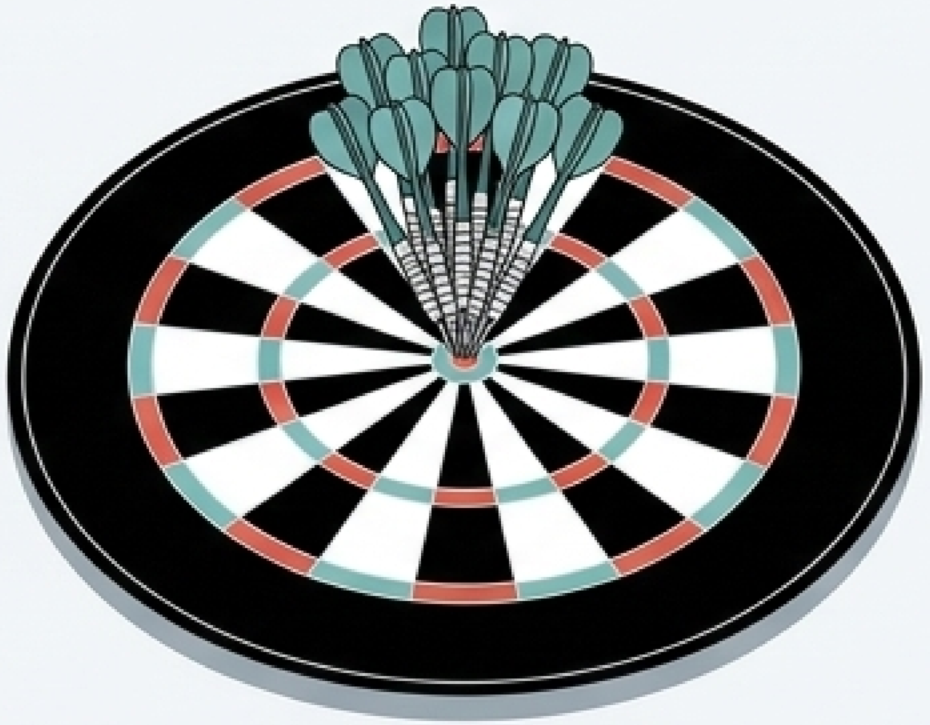
Model Misspecification

- **Excluded Variables**: A missing relevant variable leaves a systematic footprint in the error term.
- **Incorrect Functional Form**: Fitting a linear model through a marginal cost curve creates cyclical errors.

Data Issues

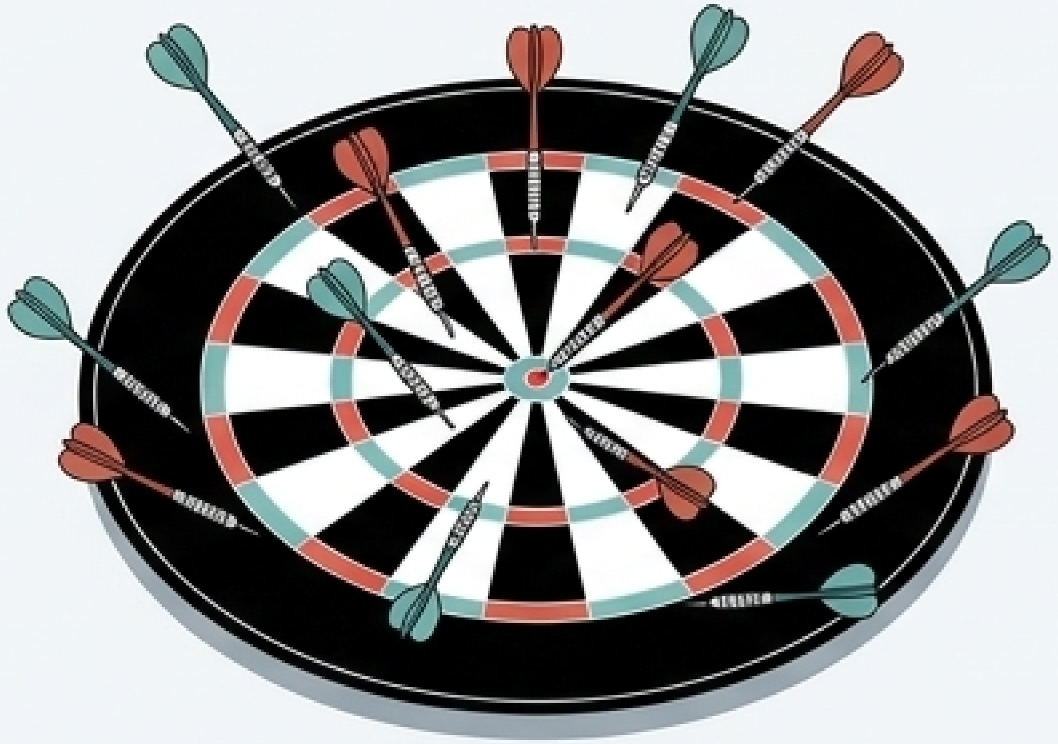
- **Manipulation**: Smoothing monthly data into quarterly averages artificially induces correlation.
- **Nonstationarity**: Time-variant means and variances within the data series.

The Consequences: The Danger of Fake Precision



Healthy OLS (BLUE)

Unbiased & Efficient. Minimum variance.



Sick OLS (Autocorrelated)

Unbiased but Inefficient. Wide variance.

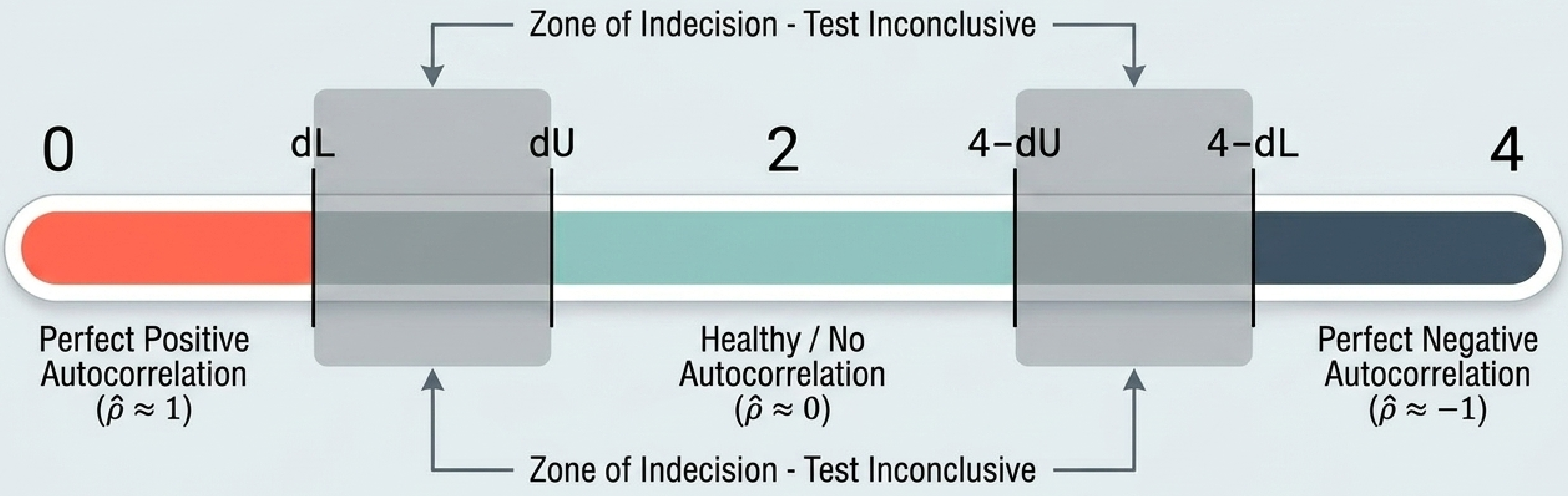
WARNING: FAKE PRECISION DETECTED

$$E(\hat{\sigma}^2) < \sigma^2$$

The residual variance underestimates the true variance. t-ratios are artificially inflated. You will declare coefficients statistically significant when they are actually not.

Diagnostic Tool 1: The Durbin-Watson d Test

$$d \approx 2(1 - \hat{\rho})$$



CRITICAL CONSTRAINTS:

- No lagged dependent variables (Y_{t-1}) allowed.
- Intercept must be present in the model.
- Only detects AR(1) errors.

Diagnostic Tool 2: The Breusch-Godfrey (BG) LM Test

The Blind Spot of Durbin-Watson

DW fails if your model includes lagged Y, or if the error is a higher-order AR(p) or Moving Average (MA) process. BG acts as the MRI to DW's stethoscope.

Step 1: Estimate

Run OLS regression on original data to obtain residuals ($\hat{u}_t$).

Step 2: Auxiliary Regression

Regress $\hat{u}_t$ on original X variables PLUS lagged residuals ($\hat{u}_{t-1}, \hat{u}_{t-2}, \ldots, \hat{u}_{t-p}$).

Step 3: Chi-Square Test

Test statistic:

$$(n-p)R^2 \sim \chi_p^2$$

TAKEAWAY:

If the computed chi-square value exceeds the critical value, reject the null hypothesis of no autocorrelation. The model is sick.

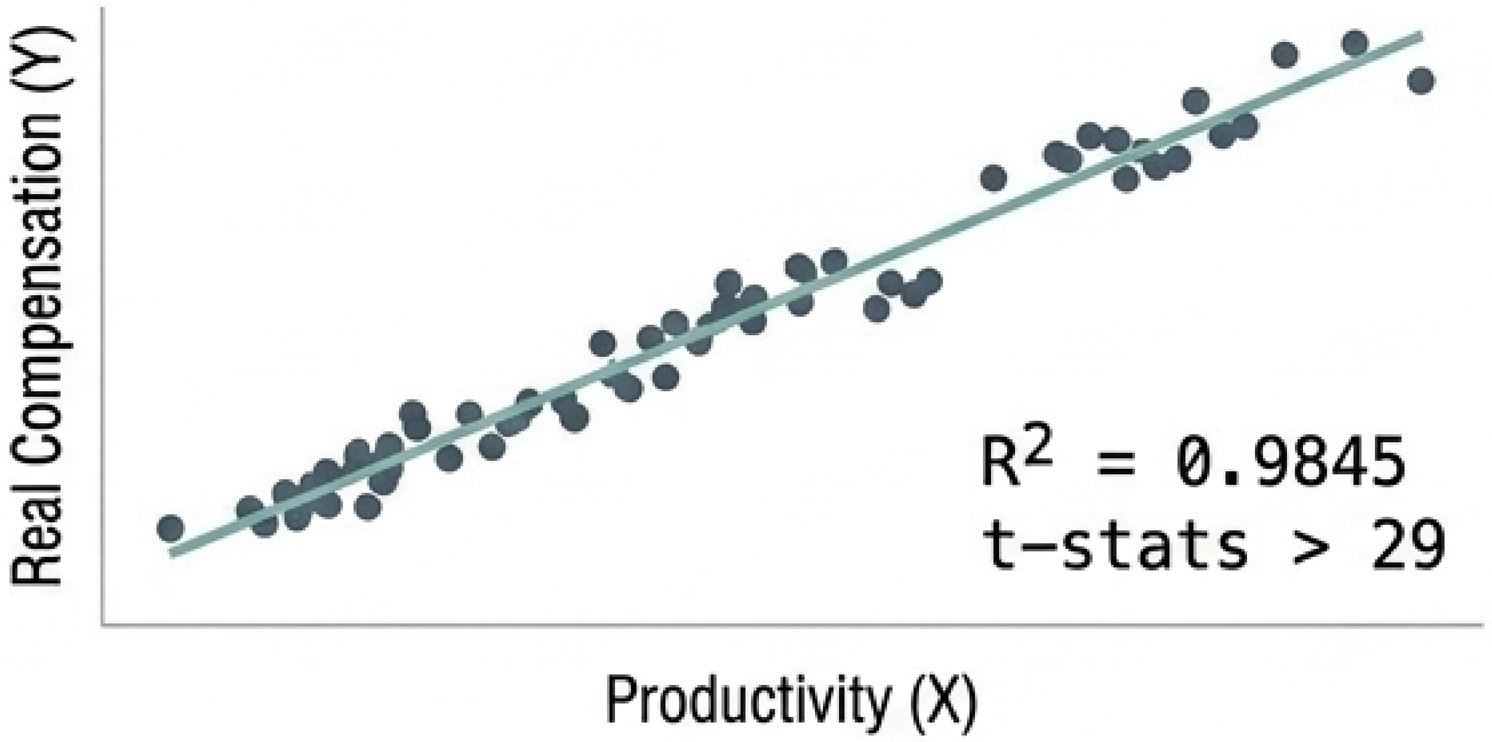
Diagnostic Showdown: DW vs. BG

Criteria	Durbin-Watson d Test	Breusch-Godfrey (LM) Test
Handles lagged dependent variables?	[X] No (Biased toward 2)	[✓] Yes
Detects higher-order AR(p) / MA?	[X] AR(1) Only	[✓] Yes, up to lag p
Requires large sample?	[✓] Works in finite small samples	[X] Asymptotic (Large sample only)
Complexity?	[✓] Automatic in all software	[X] Requires manual auxiliary regression

THE BOTTOM LINE: Use Durbin-Watson for quick checks on static, small-sample models. Prescribe Breusch-Godfrey for dynamic models or complex error structures.

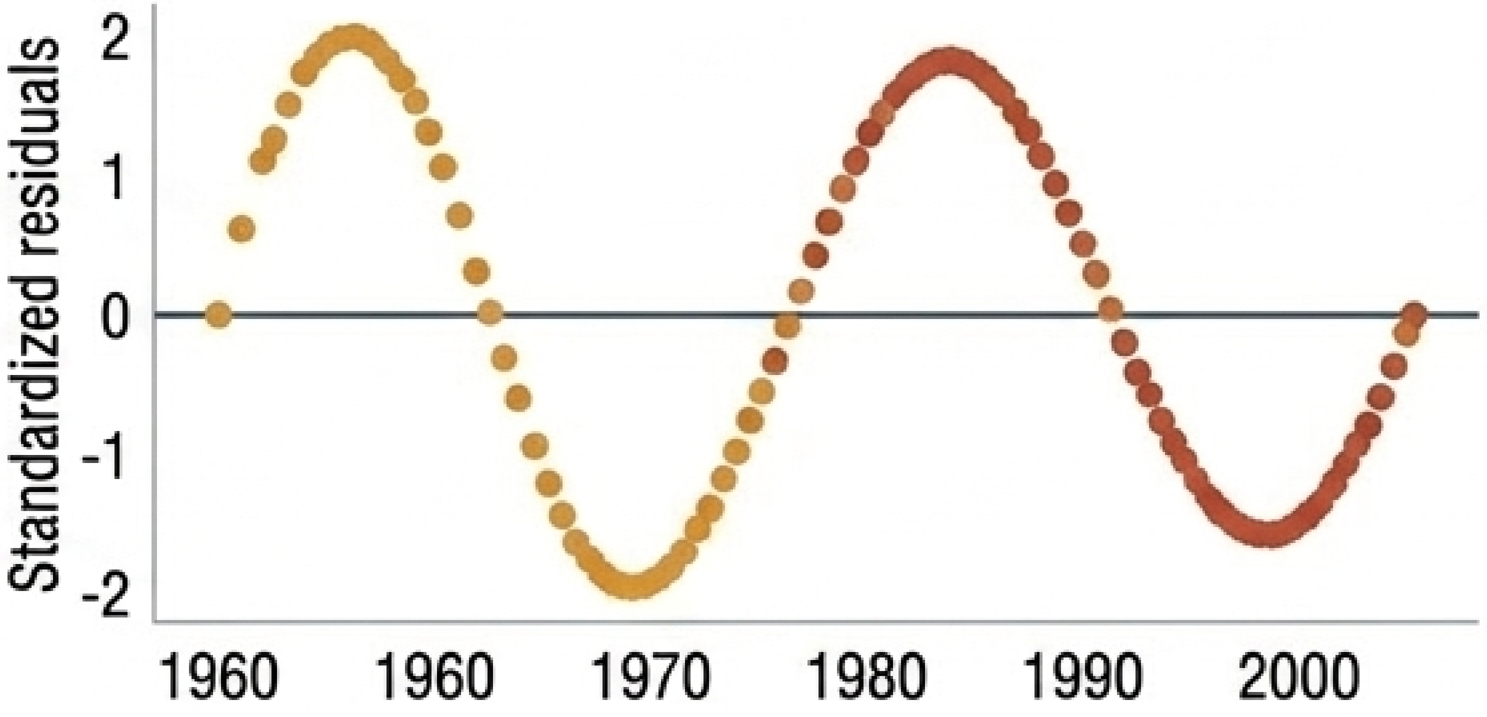
Clinical Rotation: The Wage-Productivity Case (1960-2005)

The Mirage



Looks like a perfect, highly significant model, right?

The Reality Check



Durbin-Watson d = 0.2176

CLASS WORK: Based on the d-stat of 0.2176, what is your diagnosis? Is the high t-stat trustworthy? What should the researcher do next?

The Cure, Part 1: Feasible Generalized Least Squares (FGLS)

If we know ρ, we can mathematically subtract the autocorrelated portion of yesterday from today to purify the data.

Dirty Data (Y_t, X_t)

Cochrane–Orcutt Iterative Procedure
(Estimates $\hat{\rho}$)

Purified Data
(Satisfies OLS Assumptions)

$$Y^*_t = Y_t - \rho\, Y_{t-1}$$
$$X^*_t = X_t - \rho\, X_{t-1}$$

CLINICAL WARNING: Don't lose Patient Zero! You must use the Prais-Winsten transformation on the very first observation ($Y_1 \sqrt{(1-\rho^2)}$) to avoid dropping data, which is crucial in small samples.

The Cure, Part 2: Newey-West HAC Standard Errors

Instead of surgically transforming the underlying data like FGLS, we accept the unbiased OLS coefficients but mathematically correct the Standard Errors.

$$t = \frac{\text{Coefficient}}{\text{Standard Error}}$$

HAC (Heteroscedasticity- and Autocorrelation-Consistent) standard errors automatically inflate to reflect the true variance.

BEFORE: OLS	**AFTER:** Newey-West HAC
Standard Error = 0.012 t-statistic = 52.7 (Fake Precision)	HAC Std. Error = 0.030 t-statistic = 13.7 (Realistic Precision)

[REQUIRES LARGE SAMPLE] The Newey-West procedure is asymptotic and can perform poorly in small datasets.

The Prescription Matrix:
Remedy Decision Tree

Do Nothing (OLS)

- **Best for:** Small samples where $\rho < 0.3$.
- **Adjusts Std Errors?:** No
- **Transforms Coefficients?:** No

Note: FGLS can actually perform mathematically worse than standard OLS in tiny samples with mild autocorrelation.

FGLS (Prais-Winsten)

- **Best for:** Small or Medium samples with a high ρ.
- **Adjusts Std Errors?:** Yes
- **Transforms Coefficients?:** Yes

Note: Transforms the data via Cochrane-Orcutt. Must retain the first observation to maintain efficiency.

Newey-West HAC

- **Best for:** Large samples, unknown complex error structures (handles both Autocorrelation + Heteroscedasticity).
- **Adjusts Std Errors?:** Yes
- **Transforms Coefficients?:** No

Note: Keeps the original OLS coefficients, strictly patches the standard errors.

⚠ **[SUMMARY NOTE]:** This matrix guides clinical decisions based on sample size and error structure complexity.

Final Clinical Rotation: Be the Doctor

Class Work: Read the patient charts and prescribe the correct diagnostic test or mathematical cure.

PATIENT 1

Symptoms: Model predicting GDP includes last year's GDP (Y_{t-1}) as an explanatory variable. You suspect correlation.

Prompt: Which diagnostic test do you prescribe?

[Ans: Breusch-Godfrey Test]

PATIENT 2

Symptoms: $N = 18.$ Autocorrelation is visually present in residuals. $\hat{\rho}$ is estimated at 0.85.

Prompt: Which cure do you prescribe?

[Ans: FGLS with Prais-Winsten]

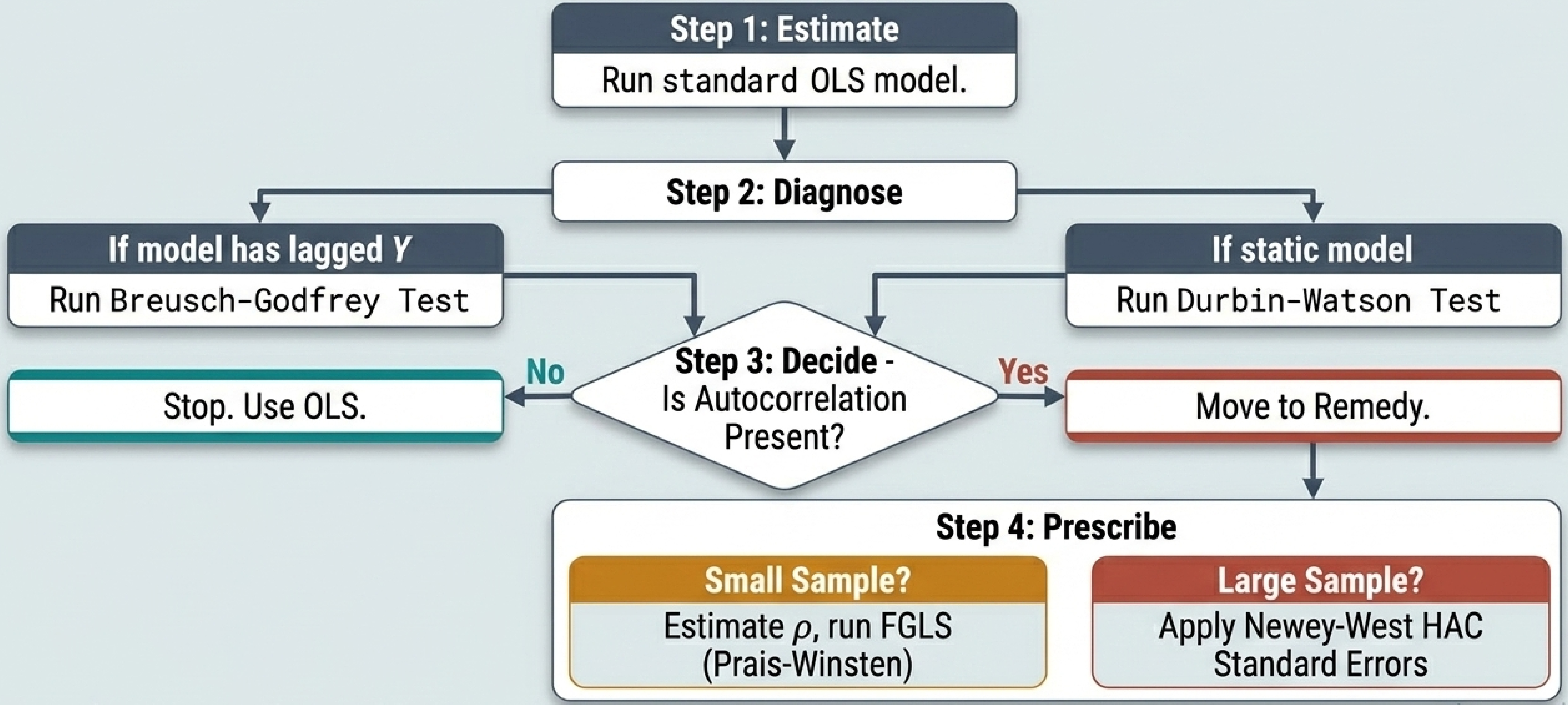
PATIENT 3

Symptoms: $N = 500.$ Residuals exhibit a complex mix of both heteroscedasticity and autocorrelation.

Prompt: Which cure do you prescribe?

[Ans: Newey-West HAC]

The Autocorrelation Action Plan



A healthy model is not one without errors, but one where errors hold no secrets.